

## COMPUTER SCIENCE MSC CURRICULUM

**Program: Software Architecture Specialization**

### ASSESSMENT TYPES:

E: Exam | PG: Practice Grade | XE: Lecture+Practice with Exam | XPG: Lecture+Practice with Practical Grade

### === COMPULSORY SUBJECTS (54 Credits) ===

#### --- SEMESTER 1 (15 Credits) ---

IPM-22RMEG: Research methodology

1 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | XPG | 5 Credits

IPM-22fesszTPE: Theory of programming (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits | Prerequisite: IPM-22fesszTPG (weak)

IPM-22fesszTPG: Theory of programming (Practice)

0 Lecture, 2 Practice, 0 Laboratory, 1 Consultation | PG | 3 Credits

IPM-22fesszPCMSG: Preparation course for master studies and developing learning skills

0 Lecture, 3 Practice, 0 Laboratory, 0 Consultation | PG | 2 Credits

IPM-22piPME: Project Management (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 0 Consultation | E | 2 Credits | Semesters: 1,3

#### --- SEMESTER 2 (28 Credits) ---

IPM-22ASTE: Advanced Software Technology (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 2 Consultation | E | 4 Credits

IPM-22DAE: Design and analysis of algorithms (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 2 Consultation | E | 4 Credits

IPM-22fesxFSE: Formal semantics (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits | Prerequisite: IPM-22fesxFSG (weak)

IPM-22fesxFSG: Formal semantics (Practice)

0 Lecture, 2 Practice, 0 Laboratory, 1 Consultation | PG | 3 Credits

IPM-22fesxDAAG: Design and analysis of algorithms (Practice)

0 Lecture, 2 Practice, 0 Laboratory, 1 Consultation | PG | 3 Credits

IPM-22fesxSQTE: Software quality and testing (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits | Prerequisite: IPM-22fesxSQTG (weak)

IPM-22fesxSQTG: Software quality and testing (Practice)

0 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | PG | 3 Credits

IPM-22fesxSALAB1: Software Technology Lab I

0 Lecture, 0 Practice, 4 Laboratory, 1 Consultation | PG | 5 Credits

#### --- SEMESTER 3 (11 Credits) ---

IPM-22fesxDDSE: Design of Distributed Systems (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits | Prerequisite: IPM-22fesxDDSG (weak)

IPM-22fesxDDSG: Design of Distributed Systems (Practice)

0 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | PG | 3 Credits

IPM-22fesxSALAB2: Software Technology Lab II

0 Lecture, 0 Practice, 4 Laboratory, 1 Consultation | PG | 5 Credits

### IPM-22PRG: Internship

0 Credits | Semesters: 2-4 | 240 hours

### === COMPULSORY ELECTIVE SUBJECTS (Select 30 Credits) ===

#### --- SEMESTER 1 ELECTIVES (15 Credits) ---

IPM-22feszAIPE: Advanced Java programming (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits | Prerequisite: IPM-22feszAIPG (weak)

IPM-22feszAIPG: Advanced Java programming (Practice)

0 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | PG | 3 Credits

IPM-22feszIDSEG: Introduction to Data Science

2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XE | 6 Credits

IPM-22feszMCE: Models of Computation (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 0 Consultation | E | 2 Credits | Prerequisite: IPM-22feszMCG (weak)

IPM-22feszMCG: Models of Computation (Practice)

0 Lecture, 2 Practice, 0 Laboratory, 1 Consultation | PG | 3 Credits

IPM-22feszPAIEG: Principles of artificial intelligence

2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XE | 6 Credits

IPM-22feszAAE: Advanced Algorithms (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits | Semesters: 1,3 | Prerequisite: IPM-22feszAAG (weak)

IPM-22feszAAG: Advanced Algorithms (Practice)

0 Lecture, 2 Practice, 0 Laboratory, 1 Consultation | PG | 3 Credits | Semesters: 1,3

#### --- SEMESTER 2 ELECTIVES (2 Credits) ---

IPM-22feszCISE: Complex information systems (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits | Prerequisite: IPM-22feszCISG (weak)

IPM-22feszCISG: Complex information systems (Practice)

0 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | PG | 3 Credits

IPM-22feszFUNLEG: Functional Languages

2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XE | 6 Credits

IPM-22fexzMLEG: Machine Learning

2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XE | 6 Credits | Prerequisite: IPM-22fexzIDSEG

IPM-22fexzSESCE: Service Science (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits | Prerequisite: IPM-22fexzSESCG (weak)

IPM-22fexzSESCG: Service Science (Practice)

0 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | PG | 3 Credits

#### --- SEMESTER 3 ELECTIVES (13 Credits) ---

IPM-22fexzAISEG: Analysis of distributed systems

2 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | XPG | 6 Credits

IPM-22fexzICSE: Introduction to Computer Security (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits | Prerequisite: IPM-22fexzICSG (weak)

IPM-22fexzICSG: Introduction to Computer Security (Practice)

0 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | PG | 3 Credits

IPM-22fexzMTAIEG: Methods and tools for AI applications

2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XE | 6 Credits

IPM-22fexzSEAPE: Scalable enterprise applications (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits | Prerequisite: IPM-22fexzSEAPG (weak)

IPM-22fexzSEAPG: Scalable enterprise applications (Practice)

0 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | PG | 3 Credits

#### === THESIS & ADDITIONAL ===

IPM-24fTHCONS: Thesis consultation

30 Credits | Semester: 4

IPM-22fERASMUS: Erasmus mobility

Maximum 24 Credits | Semester: 3

Elective Credits: 6 Credits | Semester: 3

#### === CREDIT SUMMARY ===

Compulsory Subjects: 54 Credits

Compulsory Elective Subjects: 30 Credits

Elective Subjects: 6 Credits

Thesis: 30 Credits

**TOTAL DEGREE CREDITS: 120 Credits**

Semester Credit Distribution:

Semester 1: 30 | Semester 2: 30 | Semester 3: 30 | Semester 4: 30